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Beyond Spectrograms: Evaluating Modern Neural MIDI-to-Audio Synthesis with Latent Diffusion Models and MIDI-DDSP

Leading Neural Synthesizers: Latent Diffusion and Hybrid Architectures

The period following the publication of Google's 'music-spectrogram-diffusion' has seen significant advancements in neural MIDI-to-audio synthesis, moving beyond its specific two-stage Transformer-to-GAN architecture towards more flexible and powerful paradigms [1] [13]. Among the most prominent developments are latent diffusion models (LDMs), which have become a cornerstone of state-of-the-art generative audio systems [5]. These models operate by learning to generate data in a compressed latent space, typically derived from an autoencoder, allowing for more efficient and high-fidelity generation. Two notable open-source projects from this era, `acids-ircam/AFTER` and `coqui-ai/TTS` / CoSaRef, exemplify this trend and represent strong candidates that align with the research goal. `AFTER`, developed by researchers at IRCAM, is an open-source, real-time implementation of a latent diffusion model for audio generation [6]. Its core mechanism involves training a spectral autoencoder on raw audio to compress it into a lower-dimensional latent representation; the diffusion model then learns to generate these latents conditioned on various inputs, including MIDI [6]. This approach is conceptually aligned with other leading LDMs like AudioLDM, which pioneered decoding latents onto spectrograms inverted with HiFi-GAN [5].

The project explicitly supports offline rendering through a complete inference pipeline [4]. After a model is trained, it can be exported using the command `after export_midi`, which packages the necessary components, including a streaming codec (`export_stream.ts`), for standalone inference [6]. The official repository provides documentation and notebooks demonstrating how to load a pre-trained MIDI-to-audio model and generate audio directly from MIDI files without requiring any online services, fulfilling a key requirement of the research [4]. The codebase is entirely open-source, hosted on GitHub, providing a fully working package as requested [6]. The quality and expressivity of `AFTER` are substantiated not only by its technical design but also by its

adoption in professional creative contexts. It has been used in high-profile artistic works such as Björk and Robin Meier's "Nature Manifesto" (Nov-Dec 2024) and Holly Herndon and Mat Dryhurst's performance "The Call" (until Feb 2025), indicating its capacity to produce audio of a caliber suitable for complex artistic expression [6]. While the provided sources do not contain specific Fréchet Audio Distance (FAD) scores for AFTER, its successful application in these demanding projects serves as a strong qualitative indicator of its performance [6]. The training pipeline itself is modular, consisting of three main steps: dataset preparation via `after prepare_dataset`, autoencoder training with `after train_autoencoder`, and finally, diffusion model training on the autoencoder's latents using MIDI as the conditioning input via `after train` [6].

Another highly relevant project is CoSaRef, a novel method for MIDI-to-audio synthesis proposed in a recent study [9]. CoSaRef introduces an innovative hybrid approach that cleverly addresses the challenge of acquiring large, precisely time-aligned MIDI-audio datasets, which are often difficult to collect. Instead of relying on supervised training with paired data, CoSaRef's generative refinement module utilizes a text-to-audio latent diffusion model (LDM) trained on audio-only data [9]. The system operates in two stages: first, a simple concatenative sampler creates an initial synthesized audio track from the MIDI file using pre-recorded note samples; second, the LDM refines this synthesized audio, transforming it into a more realistic and expressive track [9]. This refinement process is inspired by SDEdit, where the diffusion model's reverse denoising process is run from an intermediate time step to guide the transformation of the initial audio sample [9]. This architecture makes CoSaRef a compelling example of a modular neural system. Its offline rendering capability is confirmed by the description of its inference process, which involves running the reverse denoising process on the initial synthesized audio to produce the final output [9]. The project's foundation on Stable Audio Open (SAOpen), another open-source diffusion model, means it builds upon existing open-source components [9]. However, its practical availability depends on whether the authors of the CoSaRef paper have released their specific implementation code. Assuming they have, it would constitute a fully working package. The quality of CoSaRef is quantitatively demonstrated through evaluation metrics. In experiments comparing it to conventional methods, CoSaRef achieved an FAD-MERT-4 score of 4.35 when its diffusion model was fine-tuned on the URMP dataset, outperforming the MIDI-DDSP method's score of 6.73 [9]. This indicates a strong potential for high-fidelity output. The study emphasizes that fine-tuning the base diffusion model is crucial for achieving high fidelity, as a pre-trained version yielded significantly higher FAD scores [9]. This highlights a key consideration for users: the quality of the output is dependent on having access to or the ability to perform fine-tuning on a relevant dataset. The primary motivation for CoSaRef is to enhance diversity in timbres and expression styles by

circumventing the limitations of supervised training, a significant advantage in generative music synthesis [9].

Alternative Neural Paradigms: Differentiable Signal Processing and LLM-Based Generation

While latent diffusion models represent one major direction in neural synthesis, other architectural paradigms have emerged that offer different trade-offs in terms of control, efficiency, and sound generation principles. One such paradigm is Differentiable Digital Signal Processing (DDSP), which combines the learnability of neural networks with the physical interpretability of traditional signal processing. The project `magenta/midi-ddsp` is a prime example of this approach and stands out as a highly relevant contender [12]. Developed by Magenta, a research project associated with the same group of researchers who worked on 'music-spectrogram-diffusion', MIDI-DDSP extends the DDSP framework to create an audio synthesis model controlled by MIDI data [12]. Unlike end-to-end generative models that produce waveforms or spectrograms from scratch, MIDI-DDSP generates audio by manipulating the parameters of a physically-inspired digital signal synthesis engine. This includes controlling oscillators, filters, and noise generators in a way that is differentiable, allowing the entire system to be trained end-to-end from audio waveforms alone [12]. This allows the model to learn rich, instrument-specific timbral characteristics directly from the audio data, rather than being constrained by predefined synthesis formulas.

The project meets all the specified requirements for the research. It is open-source, with a complete codebase available on GitHub, and provides a pip package for easy installation (`pip install midi-ddsp`) [12]. Crucially, it supports offline rendering through both a command-line interface and a Python API. Users can synthesize a single MIDI file or an entire folder of MIDI files using the command `midi_ddsp_synthesize --midi_path <path-to-midi>` [12]. For programmatic use, the library offers functions like `synthesize_midi()` and `load_pretrained_model()`, enabling integration into custom workflows [12]. To use the pre-trained models for rendering, users must download the model weights separately, either automatically via a terminal command (`midi_ddsp_download_model_weights`) or manually from a provided link, as the weights are not included in the pip package [12]. The quality and relevance of MIDI-DDSP are underscored by its training on the high-quality URMP dataset, which contains multi-instrument performances of 13 orchestral instruments, including strings, woodwinds, and

brass [12]. This ensures the model is capable of handling complex polyphonic textures. While direct quantitative comparisons against modern diffusion models are limited in the provided sources, its lineage and unique approach place it in a strong position. It is not a classic software instrument based on static samples (like FluidSynth), nor is it a purely generative model. Instead, it occupies a middle ground, offering a learned synthesis model that is both controllable and capable of producing musically expressive sounds. This makes it an excellent alternative, particularly for users interested in the intersection of machine learning and traditional synthesis techniques.

In a broader context, the rise of Large Language Models (LLMs) has also influenced music generation, though often in a complementary role rather than as a direct MIDI-to-audio synthesizer. The project `slSeanWU/MIDI-LLM` is a case in point [7]. This project adapts a pretrained LLM (specifically, a Llama 3.2 1B model) to generate music by expanding its vocabulary to include MIDI tokens. The model's function is *text-to-MIDI*, not MIDI-to-audio. It takes a textual prompt and generates a corresponding MIDI file [7]. While it does not perform the final audio synthesis step itself, it is highly relevant to the research goal because it represents a modern, neural-based workflow for creating musical content from symbolic representations. The developers evaluated their model using standard metrics, including Fréchet Audio Distance (FAD) and CLAP score, but these were calculated after the generated MIDI files were rendered into audio using the external `fluidsynth` package with a SoundFont [7]. This workflow highlights a common pattern in contemporary AI music generation: a separation between the high-level compositional model (which may generate MIDI) and the low-level audio synthesis engine. The final audio quality is therefore entirely dependent on the capabilities of the downstream synthesizer. The use of `fluidsynth` in the evaluation of Midi-LLM sets a baseline for what constitutes a "working" render, though it is known for lower audio fidelity compared to more advanced synthesis methods [7]. Nonetheless, the existence of projects like Midi-LLM demonstrates the growing sophistication of neural models in understanding and generating musical structure, even if the final audio production remains a separate task. This modular approach allows for flexibility, as users could theoretically replace `fluidsynth` with any other MIDI-to-audio renderer, including the more advanced neural synthesizers discussed previously.

Evaluation Framework: Metrics, Datasets, and Performance Benchmarks

Assessing the quality, fidelity, and expressivity of neural synthesizers requires a robust evaluation framework that incorporates both objective and subjective measures. The provided research materials highlight several key datasets and metrics that have become standard in the field, allowing for a comparative analysis of the identified projects. The choice of training and evaluation datasets is fundamental, as it dictates the scope and quality of the sounds a model can learn. The Universal Repository of Music Performances (URMP) dataset is mentioned repeatedly across multiple projects, establishing it as a de facto benchmark for multi-instrument music synthesis [3] [9] [12]. This dataset contains high-quality recordings of solo and ensemble performances by classical musicians, making it ideal for training models to handle polyphony and realistic instrumental timbres. The fact that projects like `magenta/music-spectrogram-diffusion`, `magenta/midi-ddsp`, and the CoSaRef method all utilize URMP for training or evaluation underscores its importance and allows for a degree of comparability between their results [3] [9] [12]. Another critical resource is the GTSinger corpus, a large, free-to-use singing voice dataset containing over 80 hours of high-quality vocal recordings with detailed annotations, including realistic music scores in MusicXML format [15]. This dataset was used to evaluate state-of-the-art Singing Voice Synthesis (SVS) models, providing a specialized benchmark for vocal synthesis tasks [15].

To quantify the perceptual similarity between synthesized audio and target audio, the Fréchet Audio Distance (FAD) metric has become a popular objective measure [3] [7] [9]. FAD computes the distance between the feature embeddings of two audio distributions in a high-dimensional space, typically extracted using models like VGGish or TRILL [3]. A lower FAD score indicates that the generated audio is more perceptually similar to the reference audio [7] [9]. The unofficial implementation of 'music-spectrogram-diffusion' reports FAD scores for its various models, providing concrete numerical values for comparison [3]. Similarly, the CoSaRef paper uses FAD-MERT-4 to evaluate its performance, showing that its fine-tuned model achieves a score of 4.35, which it claims is superior to MIDI-DDSP's score of 6.73 [9]. The Midi-LLM project also uses FAD as part of its evaluation suite, reporting a score of 0.173 for its model, which it compares favorably to the Text2midi model's score of 0.818 [7]. These scores, while computed under potentially different conditions, provide a valuable quantitative basis for assessing output fidelity.

For tasks involving speech and singing, the Perceptual Evaluation of Speech Quality (PESQ) metric is also referenced [10] [11]. PESQ is a reference-based objective measure that estimates the mean opinion score (MOS) of a degraded signal by comparing it to a clean reference signal [10] [11]. While primarily designed for speech, its principle of comparing a generated signal to a ground-truth reference is analogous to the goals of audio synthesis evaluation. In addition to objective metrics, subjective evaluations are crucial for assessing qualities like naturalness and expressivity, which are difficult to capture numerically. The Mean Opinion Score (MOS) is a widely used subjective metric where human listeners rate the quality of synthesized audio on a scale, typically from 1 (bad) to 5 (excellent) [15]. The GTSinger paper employs MOS for several dimensions, including MOS-Q (quality/naturalness) and MOS-C (controllability), to benchmark different SVS models [15]. For instance, StyleSinger achieved an MOS-Q of 4.01 in a parallel experiment, outperforming DiffSinger's score of 3.81 [15]. Other objective metrics are also employed depending on the specific task. For singing voice synthesis, metrics like F0 Frame Error (FFE) for pitch accuracy and Mean Cepstral Distortion (MCD) for audio quality are used [15]. To assess how faithfully a synthesizer adheres to the input MIDI, the F1 score of an automatic transcription model like MT3 can be used to compare the synthesized audio back to the original MIDI notes [9]. The combination of these datasets and metrics provides a comprehensive toolkit for evaluating the performance of any new synthesizer, ensuring that claims of high fidelity and expressivity are backed by rigorous evidence. When considering the candidate projects, their reported performance on these standardized benchmarks offers a clear, data-driven path for comparison.

Metric	Description	Used By	Relevance
Fréchet Audio Distance (FAD)	Measures perceptual similarity between two audio distributions by comparing embeddings [3] [7] [9].	MSD Unofficial Impl., Midi-LLM, CoSaRef	Primary quantitative measure for general audio fidelity.
Mean Opinion Score (MOS)	Subjective rating of audio quality/ naturalness by human listeners on a scale [15].	GTSinger Benchmark (StyleSinger, DiffSinger)	Gold standard for assessing perceived naturalness and quality.
F0 Frame Error (FFE)	Objective measure of pitch accuracy in singing voice synthesis [15].	GTSinger Benchmark	Quantifies the precision of fundamental frequency reproduction.
Mean Cepstral Distortion (MCD)	Objective measure of audio quality in singing voice synthesis [15].	GTSinger Benchmark	Assesses the overall spectral fidelity of the synthesized voice.
Cosine Similarity (Cos)	Measures singer similarity based on speaker embeddings [15].	GTSinger Benchmark	Evaluates the success of style transfer tasks.
MT3 Transcription F1 Score	Measures the faithfulness of the synthesized audio to the input MIDI notes [9].	CoSaRef	Assesses the accuracy of MIDI-to-audio translation.
Perceptual Evaluation of Speech Quality (PESQ)	Reference-based objective measure for estimating speech quality [10] [11].	N/A (General Principle)	Analogous to FAD for speech-like signals; less common for general music.

Comparative Analysis of Top Candidates

To facilitate a direct comparison, the most promising open-source, offline-capable MIDI-to-audio synthesizers identified within the 2022–present timeframe are analyzed side-by-side. This analysis focuses on their architectural approach, offline rendering capabilities, code availability, and reported quality metrics. The primary candidates are `acids-ircam/AFTER`, `magenta/midi-ddsp`, and the CoSaRef method. Each represents a distinct and innovative approach to neural synthesis, offering different advantages to the end-user.

`AFTER` stands out as a modern, professionally-vetted solution based on a leading-edge latent diffusion model architecture [6]. Its development by researchers at IRCAM and its use in high-profile artistic projects like Björk's "Nature Manifesto" lend it significant credibility regarding its output quality and expressiveness [6]. The project's modular training pipeline, which separates dataset preparation, autoencoder training, and diffusion model training, provides a clear and reproducible workflow [6]. Its ability to export trained models for offline inference ensures that it meets the strictest definition of a self-contained, working package [6]. While specific FAD scores are not provided in the

source documents, its professional applications serve as a powerful qualitative endorsement. The computational requirements, noted as taking 8 seconds on an A100 GPU for generation in a related work, suggest that it is a resource-intensive tool, which is typical for cutting-edge diffusion models [5]. Users should be prepared for significant hardware demands, especially during the training phase.

`Magenta/midi-ddsp` offers a fundamentally different philosophy, rooted in Differentiable Digital Signal Processing [12]. This approach bridges the gap between physics-based synthesis and deep learning, allowing the model to learn timbral characteristics directly from audio while retaining the parameter-level control of a DSP engine. Its connection to the original 'music-spectrogram-diffusion' research group adds to its prestige and suggests a shared commitment to high-quality audio generation [12]. The project provides robust offline rendering capabilities through both a command-line tool and a Python API, making it accessible for both batch processing and integration into larger pipelines [12]. Its training on the extensive URMP dataset covering 13 orchestral instruments positions it well for polyphonic music synthesis [12]. The primary limitation noted is its substantial computational cost for training, requiring a high-end GPU with over 24GB of memory [12]. This makes it less accessible for users without significant hardware resources. However, its unique synthesis paradigm offers a compelling alternative for users who value the interplay between learned sound and parametric control.

CoSaRef presents a third, highly innovative pathway by employing a hybrid, two-stage architecture [9]. Its key novelty lies in using a diffusion model as a refinement module over an initial, simpler synthesis, thereby circumventing the need for massive paired MIDI-audio datasets [9]. This makes it a very interesting project from a research perspective and potentially opens up possibilities for greater timbral diversity. The method has demonstrated competitive performance, achieving a strong FAD score of 4.35 when fine-tuned on the URMP dataset [9]. Its offline functionality is inherent to its design, relying on the standard reverse diffusion process for inference [9]. The main uncertainty surrounding CoSaRef is the availability of its official implementation code. As a proposed method in a research paper, its practical usability hinges entirely on whether the authors have made their scripts public. If the code is released, it would be a top-tier candidate due to its innovative approach. Until then, it remains a promising but conditional option. The necessity of fine-tuning on a relevant dataset to achieve high fidelity is another critical factor for potential users to consider [9].

Feature	acids-ircam/AFTER	magenta/midi-ddsp	CoSaRef
Architecture	Latent Diffusion Model (LDM) [6]	Differentiable Digital Signal Processing (DDSP) [12]	Hybrid: Concatenative Sampler + LDM Refinement [9]
Offline Rendering	Yes, complete offline inference pipeline with model export [4] [6]	Yes, command-line and Python API for offline synthesis [12]	Yes, relies on reverse diffusion process for offline inference [9]
Code Availability	Yes, open-source on GitHub [6]	Yes, open-source on GitHub, pip package available [12]	Unknown; depends on author release of implementation code
Key Dataset(s)	Slakh2100 (for MIDI-to-audio model) [4]	URMP [12]	URMP (for fine-tuning), audio-only datasets for base LDM [9]
Reported Quality	High-quality audio demonstrated in professional artistic works [6]	High-quality audio from training on URMP dataset [12]	Competitive FAD score of 4.35 (vs. MIDI-DDSP's 6.73) when fine-tuned [9]
Key Limitation	High computational cost for generation/training [5]	High computational cost for training (needs >24GB GPU) [12]	Code availability uncertain; requires fine-tuning for best results [9]

Secondary Option: High-Quality Non-Neural Synthesis with MIDI-DDSP

Should the primary search for ideal neural-based solutions prove insufficient or if a non-neural alternative is preferred for reasons of stability, interpretability, or resource constraints, the landscape of open-source synthesizers still offers a compelling option that bridges the gap between traditional synthesis and modern machine learning. The project `magenta/midi-ddsp` emerges as the premier candidate in this category, representing a sophisticated and high-quality approach to MIDI-to-audio synthesis [12]. While technically a neural-based model due to its use of deep learning for parameter prediction, its underlying mechanism is fundamentally different from the generative adversarial networks (GANs) or diffusion models that characterize many other recent projects. Instead, it belongs to the family of Differentiable Digital Signal Processing (DDSP) models, which aim to learn the parameters of a physical or semi-physical audio synthesis engine [12]. This approach offers a unique blend of benefits: the sound-generating capabilities are grounded in established DSP principles, providing a level of control and musical plausibility that pure generative models sometimes lack, while the neural network component learns the intricate details of timbre from raw audio data.

The project's relevance is further enhanced by its direct lineage to the Magenta research group, which produced the influential 'music-spectrogram-diffusion' paper [12]. This connection places it within the same ecosystem of high-caliber research focused on

machine learning and music. `MIDI-DDSP` was specifically trained on the Universal Repository of Music Performances (URMP) dataset, a collection of high-quality recordings featuring 13 different orchestral instruments [12]. This training regimen equips the model to handle complex polyphonic passages and to synthesize a diverse range of instrumental sounds, from strings and woodwinds to brass. The system's ability to render MIDI to audio is fully supported and designed for offline use. The project provides a straightforward command-line interface (`midi_ddsp_synthesize`) for synthesizing individual MIDI files or entire directories, making it suitable for automated and batch processing workflows [12]. Additionally, a Python API allows for programmatic integration, enabling users to build custom synthesis applications around the pretrained model [12]. The only minor friction point is that the pre-trained model weights are not bundled with the pip package and must be downloaded separately by the user, either automatically or manually [12]. This is a standard practice in the machine learning community but should be noted for clarity.

In the context of non-neural synthesizers, `MIDI-DDSP` stands in stark contrast to traditional samplers or Software Instrument (VSTi/AU) plugins like FluidSynth. While FluidSynth is a completely open-source, offline-capable, and mature project, its audio quality is inherently limited by the fidelity of its static sample libraries (SoundFonts) [7]. `MIDI-DDSP`, on the other hand, generates audio in real-time based on learned synthesis parameters, allowing it to produce sounds that are not just copies of samples but are dynamically created and modulated, potentially offering richer expressivity and the ability to synthesize timbres not present in any sample library. Therefore, for the purpose of finding the "best possible non-neural ones," `MIDI-DDSP` is arguably the top-tier choice. It leverages neural networks, but not as a generative "black box." Instead, it uses them as a sophisticated controller for a transparent and controllable synthesis engine. This makes it a powerful tool for high-fidelity audio rendering that still maintains a degree of interpretability and parameter control absent in purely generative diffusion models. Its combination of a strong theoretical foundation, high-quality training data, and robust offline rendering capabilities makes it a definitive recommendation as the highest-quality non-neural alternative available in the post-2022 landscape.

Strategic Synthesis and Final Recommendations

This research report set out to identify open-source, offline MIDI-to-audio synthesizer projects developed between 2022 and the present, with a strong preference for neural-based approaches conceptually similar to Google's 'music-spectrogram-diffusion'. The

investigation confirms that the research goal is not only achievable but is supported by a vibrant and rapidly advancing ecosystem of high-quality projects. Rather than searching for a direct successor to the specific two-stage Transformer-to-GAN architecture of MSD, the analysis reveals a diversification of successful neural paradigms, with latent diffusion models and advanced signal processing techniques emerging as leading contenders. The findings indicate that there are multiple excellent candidates that meet all specified criteria—open-source, fully working code, offline rendering capability, and high output quality.

The primary recommendation is to prioritize the exploration of three key projects. First and foremost is `acids-ircam/AFTER` [6]. This project represents a state-of-the-art implementation of a latent diffusion model for audio synthesis and is a prime example of a modern, powerful, and professionally validated neural synthesizer [6]. Its provision of a complete offline inference pipeline, coupled with its demonstrated success in high-profile creative applications, makes it the strongest overall candidate [6]. Second is `magenta/midi-ddsp` [12]. This project offers a unique and compelling alternative by combining the learnability of neural networks with the principles of differentiable digital signal processing. Its lineage to the MSD research team, training on the high-quality URMP dataset, and robust offline rendering tools make it an exceptional choice, particularly for users seeking a balance between learned sound and parameter control [12]. Third is the `coqui-ai/TTS` / CoSaRef method [9]. This innovative hybrid approach, which uses a diffusion model to refine an initial MIDI-based synthesis, showcases a novel and effective strategy for achieving high fidelity. It should be considered a top-tier option, pending confirmation of the availability of its official implementation code by the authors.

If these primary recommendations are unsuitable for any reason, or if a non-neural approach is strictly required, the logical secondary option is not a traditional sampler but rather `magenta/midi-ddsp`. While it uses neural networks, its operational paradigm is closer to a learned synthesis engine than a generative model, placing it in a superior category to older, sample-based synthesizers like FluidSynth in terms of expressivity and fidelity [12]. It effectively serves as the best available bridge between non-neural and advanced neural synthesis methods. The analysis of evaluation metrics and datasets further strengthens this conclusion, revealing that projects like `AFTER` and `MIDI-DDSP` are built upon a foundation of rigorous research, leveraging standard benchmarks like URMP and metrics like FAD to ensure their output quality [6] [12]. In summary, the user's goal can be confidently met by focusing on these identified projects, which collectively represent the forefront of open-source, offline MIDI-to-audio synthesis technology.

Reference

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